

Practical Exam – Automotive reliability

From:	Head of Data Science
Received:	Today
Subject:	New project request from the Fleet Reliability Team
Hi, I have a new project for you from the Fleet Reliability Team. It should be a practical, high-impact piece of work. You can see the background and request in the email below. I would like you to perform the analysis and write a short report for me. I want to be able to review your code as well as read your thought process for each step. I also want you to prepare and deliver the presentation for the Fleet Reliability team - you are ready for the challenge! The team needs to reduce scooter downtime while keeping maintenance and parts costs under control, and wants us to identify the strongest drivers of scooters going out of service. I'm not sure if that's possible with the data quality and time window, but do your best and present whatever you find. You can find more details about what I expect you to do here. And information on the data here. I will be on vacation for the next couple of weeks, but I know you can do this without my support. If you need to make any decisions, include them in your work and I will review them when I am back. Good luck!	

From:	Head of Strategy
To:	Head of Data Science
Received:	Yesterday
Subject:	Reduce scooter downtime with proactive maintenance insights
Hi, I'm reaching out from the Fleet Reliability Team with an analysis request: Over the last few months we've seen an increase in scooters becoming unavailable during peak periods, and we need to understand why and what we can do about it. At a high level, the problem is that scooters are intermittently going out of service (battery issues, mechanical faults, and rider-reported safety concerns). Our operations team is reacting after issues occur rather than preventing them, which creates avoidable downtime and increases support tickets. Can your team help us to: • Identify the most influential predictors of a scooter being taken out of service in the 24 hour period after the snapshot • Given these data, predict whether a scooter will go out of service in the following 24 hour period with at least 90% accuracy. We are making staffing and purchasing decisions soon, so I need you to present the results and recommendations at our next Reliability Review. Whatever your results, what do you recommend we do next? Look forward to seeing your presentation. Thanks.	

About the Company

QuorWatt Urban Mobility is a shared micromobility operator that deploys dockless electric scooters in multiple cities. Riders locate scooters using a mobile app, unlock them for short trips, and end rides in permitted parking zones.

The Fleet Reliability Team owns fleet availability, safety, and downtime reduction. Their success is measured by scooter availability during peak times, fewer safety-related incidents, and efficient use of technician labor and replacement parts.

Scooter reporting information

Each scooter periodically sends telemetry to QuorWatt systems (battery status, basic health pings, and ride counters). The mobile app allows riders to submit issues, which are logged against a scooter. Operational systems also record when a scooter is marked unavailable (out of service) and when it returns to service.

Each row in the dataset is a scooter-level snapshot. Features summarize the prior 24 hours leading up to the snapshot. The target indicates whether the scooter goes out of service within the following 24 hours.

Data Information

Our research team has collected the dataset below. The dataset contains scooter-level snapshots. Each row represents one scooter, with features summarizing the prior 24 hours and an outcome describing whether maintenance was required shortly afterward.

You can find the data [here](#), as well as inside the workbook template for this task.

I will let you decide how to process it, just make sure you include all your decisions in your report.

Don't forget to double check the data really does match what they say - it might not.

Column Name	Details
scooter_id	Discrete. Unique identifier for the scooter unit.
service_area	Categorical. Operational zone (city/area) where the scooter was primarily observed during the snapshot period.

scooter_model	Categorical. Scooter hardware model/family identifier.
total_trips_24h	Discrete. Number of completed trips recorded in the prior 24 hours
battery_health_score	Continuous. Battery health estimate on a 0–100 scale (higher is healthier).
reported_issue_count_24h	Discrete. Number of rider-reported issues logged in the prior 24 hours
taken_out_of_service	Nominal. 1 if the scooter was taken out of service for repair in the following 24 hours; else 0.

Guide to Data Science Projects

1. I would like you to create a written report to summarize the analysis you have performed and your findings. The report will be read by me (Head of Data Science). The list below describes what I expect to see in your written report.
2. You will need to use a DataLab workbook to complete your analysis, write up your findings and share visualizations.
3. You must use the data provided for the analysis.
4. You will also need to prepare and deliver a presentation. You should prepare around 6–10 slides to present. The list below describes what they expect to see in your presentation.
5. Your presentation should be no longer than 10 minutes.

Written Report

Your written report should include written text summaries and graphics of the following:

- Data validation:
 - Describe validation and cleaning steps for every column in the data
- Exploratory Analysis to answer the customer questions ensuring you include:
 - Two different types of graphic showing single variables only
 - At least one graphic showing two or more variables
 - Description of your findings
- Model Development including:
 - What type of problem this is

- Fitting a baseline model
 - Fitting a comparison model
- Model evaluation
 - Show how the two models compare
- Definition of a metric for the business to monitor
 - How should the business monitor what they want to achieve?
 - Estimate the initial value(s) for the metric based on the current data?
- Final summary including recommendations that the business should undertake

Presentation

You will give an overview presentation to the individual who requested the work. The presentation should include:

- An overview of the project and business goals
- A summary of the work you undertook and how this addresses the problem
- Your key findings including the metric to monitor and current estimation
- Your recommendations to the business

Grading

Before submitting your written report or delivering your presentation, remember to check your work against the grading criteria.

You can find the full grading rubric [here](#).

You must pass all criteria to pass this part of the certification.